ALHASSAN JAWAD

SOFTWARE DEVELOPER • FOCUS: CYBER- & OT-SECURITY

+46 76 281 8440 | alhassan0000@yahoo.com | Stockholm, Sweden
www.alhassanjawad.com | github.com/AcheronEiden | linkedin.com/in/alhassanjawad

PROFESSIONAL SUMMARY

Software Engineer currently working part-time at Abbott Laboratories while simultaneously completing a M.Sc. in Electrical Engineering (expected Q1 2027) with coursework centered on **OT-security** and **programming of embedded systems**. Proven experience delivering production features across frontend frameworks, and backend databases.

EDUCATION

Master of Science (M.Sc.) in Electrical Engineering
2025 – 2027

Uppsala University – Specialization: Control and Programming of Embedded Systems

Bachelor of Science (B.Sc.) in Electrical Engineering
2019 – 2022

Department of Electrical Engineering | Uppsala University
Degree issued in 2026

TECHNICAL COMPETENCIES

Programming Languages: Python, C/C++, Java, MATLAB, JavaScript, TypeScript, HTML5, CSS, SQL (PostgreSQL)

Engineering Tools & Frameworks: Git/GitHub, Angular, Node.js, Tailwind, LabVIEW

Languages: Swedish (Fluent), English (Fluent), Arabic (Native)

CERTIFICATIONS

- **micro1:** Vetted Software Engineer (Electronic Engineering / Ngspice) [Issued 2026]
- **Google Cloud Certified**: Professional Google AI Engineer [Issued 2026]

RELEVANT EXPERIENCE

IT Infrastructure Specialist - Abbott Laboratories
Sep 2026 - Ongoing

Maintain high-availability endpoint configurations across Nordic facilities (Sweden, Denmark, Finland) as the Tier-3 escalation contact. Program automated PowerShell and batch routines to handle client imaging, driver installation, and application deployments. Query and audit Active Directory/LDAP services, debugging endpoint authorization faults, script errors, and network socket communication issues.

Software Engineer (Internship) – Portabel Health
Sep 2025 – Dec 2025

- Developed and deployed client-facing platform features using Angular, TypeScript, and Tailwind CSS within an Agile workflow.
- Executed 130+ development and bug-fix tasks tracked via Freedcamp across staging and production branches.
- Structured relational queries, table migrations, and schema optimizations using PostgreSQL.
- Established standardized internal Git version control guidelines and authored refactoring documentation for codebase maintainability.

Teaching Assistant – Uppsala University
Aug 2023 – Jul 2025

- Instructed and evaluated 50+ students per semester on object-oriented programming concepts, control structures, and debugging techniques.
- Conducted structured code reviews and graded laboratory assignments in introductory C and Python courses.

SELECTED PROJECTS

Master's Thesis: Automated Aeroponic Cultivation System 2025

Internal Collaboration with Uppsala University

- Programmed multi-variable MIMO feedback control algorithms in Embedded C on an ESP32 dual-core microcontroller.
- Architected a non-blocking I2C sensor polling state machine to stream environmental metrics into real-time telemetry routines.
- Handled memory optimization, hardware timer interrupts, and register configurations for reliable multi-sensor execution.

Bachelor's Thesis: Smart Building Energy Management 2022

External Collaboration with Studenternas IP, Uppsala

- Wrote numerical optimization algorithms in MATLAB to process time-series electrical load data.
- Processed multi-variable datasets from facility power meters, isolating 71.2 kW out of a 185 kW peak load as manageable power flexibility.
- Proved that 38.5% of the infrastructure's peak demand could be shifted to reduce peak grid tariffs without affecting baseline operations.

Project: Miniature Elevator Prototype (Course: Project in Electronics 10 ECTS) 2023

Academic Project | Uppsala University

- Implemented an event-driven state machine architecture.
- Programmed the control logic and GUI (in LabVIEW), to manage real-time level selection and state transitions while regulating motor speed profiles, to achieve a smooth acceleration while minimizing physical oscillations.

Project: GPS Tracking Device (Course: PC-Board Construction with ECAD Tools 10 ECTS) 2022

Academic Project | Uppsala University

- Developed C firmware for an ATmega328P MCU interfacing with a Quectel MC60 GSM/GPS transceiver via hardware UART.
- Built a custom NMEA sentence parser to extract GPS coordinates and packaged telemetry into TCP/GPRS payloads for transmission.
- Implemented low-power sleep modes triggered by hardware interrupt routines from an MMA8653 accelerometer.

References

Available upon request

Thank you for taking the time to look at my CV
If you have any questions, don’t hesitate to contact me @ alhassanjawad.com/#contact